

Northwind Traders - Executive Summary

Data Analytics and Visualization Course Project | PostgreSQL, DBeaver and Microsoft Power BI

830 Orders	2,155 Sales Lines	77 Products	89 Active Customers	EUR 1.266M Revenue
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Project Purpose

The project delivers an end-to-end business intelligence solution for Northwind Traders. The goal was to transform the original relational database into a reusable analytical model and an interactive Power BI report that supports analysis of sales performance, products, customers, employees and shipping activity.

Approach and Architecture

The solution was built in layers so each technology has a clear responsibility. PostgreSQL was used for exploration and structural data preparation, Power Query only for lightweight cleanup, Power BI for semantic modeling, DAX for reusable business calculations, and report pages for presentation and interaction. This separation keeps the solution easier to validate, maintain and extend.

SQL and Data Preparation

Three reusable analytical views were created: **v_Product_Enriched**, **v_Order_Sales** and **v_Customer_Geo**. The main fact view was intentionally kept at order-line grain so that product-level analysis and drill-through remain possible while higher-level measures aggregate correctly. Revenue was calculated in SQL as **Unit Price x Quantity x (1 - Discount)**. Power Query then handled data types, readable field names and removal of unnecessary navigation columns.

Data Model and DAX

Power BI uses a **Star Schema** with **v_order_sales** as the central fact table and Product, Customer, Employee, Shipper and Calendar dimensions around it. Relationships are active and single-directional. A dedicated Calendar table supports time intelligence, while a separate **_Measures** table stores Total Revenue, Total Orders, Average Order Value, Total Quantity Sold, Revenue YTD, Previous-Year Revenue, YoY Growth, discounted revenue share and Top Product Sales.

Dashboard and Interaction

The final report contains three complementary pages. **Executive Overview** provides headline KPIs, monthly trends, geographic sales distribution and shipper usage. **Product & Employee Analysis** supports deeper investigation through product rankings, employee performance, slicers and a Category -> Product -> Customer decomposition tree. **Order Details** is a drill-through page that exposes the underlying transactions for a selected product. Reset-filter bookmarks, cross-filtering and Back navigation improve usability.

Outcome

The final solution demonstrates a complete analytics workflow from relational source data to validated business insight. The chosen design separates data preparation, business logic and presentation, producing a report that is clear, reusable, maintainable and easy to validate against the source data.